Venkata Lakshmi Parimala Pasupuleti

Fullstack Software Engineer · PHP · Python · React · TypeScript · Compliance & Security · Distributed Systems · Audit Logging · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Fullstack software engineer with 9+ years of production experience building secure, compliant, and highly available enterprise SaaS systems - the reliability and compliance standards Slack's Enterprise Trust team requires at Fortune 100 scale. At Entrata, I owned mission-critical compliance infrastructure for 9 years: PHP and Python backend systems serving 100K+ housing units with 100% PHI/HIPAA compliance, Libsodium-encrypted data pipelines, full audit logging across every regulated action, role-based access controls, and zero security incidents across the entire ownership period. I shipped 20+ full-stack production modules using PHP, Python, PostgreSQL, JavaScript, and React across 120+ Agile sprints - the same multi-language, product-driven engineering model the Enterprise Trust team uses. My compliance background goes beyond checkbox implementation: I owned the annual HUD federal compliance pipeline where a single incorrect data record triggers incorrect eligibility decisions and regulatory violations for thousands of residents - the same stakes Slack's enterprise customers face when SSO, data retention, or audit log systems fail. I build systems where correctness and security are non-negotiable, not afterthoughts. I also integrate AI tools (LeaseI, AgentProbe, Claude Code) to improve engineering velocity while maintaining compliance and auditability. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Fullstack: Python (9yr), PHP (2yr production SaaS), TypeScript, JavaScript, React.js, SQL, PL/SQL, Bash, HTML/CSS, jQuery, FastAPI, Flask, REST APIs, Linux (production 9yr), Git

Security & Compliance: PHI/HIPAA Compliance, Data Security & Encryption (Libsodium), Audit Logging, RBAC, Access Controls, Secure Application Design, Data Governance, Privacy-by-Design, Regulatory Compliance, Data Retention, Compliance Pipelines, Zero Security Incidents 7yr

Infrastructure & Data: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (event-driven async), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Automated Testing, Performance Engineering, Distributed Systems, Observability, Agile/Scrum, LangGraph, LangChain, AI/ML Integration

PROJECTS

LeaseI: Compliance AI System - Audit Logging, RBAC & Data Governance

Compliance · Audit Logging · Security · HITL · Production

Python · LangGraph · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production compliance system built with enterprise-grade security and auditability - the same design standards Slack's Enterprise Trust team applies to regulated enterprise workflows. **Audit logging:** complete decision trail across all actions - every regulated decision is logged with timestamp, source, agent, confidence score, and outcome; immutable audit records for compliance review. **Data governance:** encrypted data handling throughout; data minimization patterns; deterministic guardrails ensure AI never auto-resolves compliance decisions without human approval. **RBAC:** role-based access controls with full data isolation. 93.6% precision across 1,240 compliance test cases; CI regression gate blocks quality regressions before deployment.

AgentProbe: Secure AI Platform - Access Controls & Observability

Security · Observability · API Design · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production AI platform with security, observability, and API design as first-class requirements - 15-endpoint FastAPI backend with clean API abstractions; secrets management and access controls for multi-provider LLM routing (Anthropic, OpenAI, Ollama); 5-metric observability engine with execution tracing and audit logging; CI-integrated regression gate. Designed with the same secure-by-default, observable-by-default principles enterprise compliance systems require. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

AI on FHIR: HIPAA-Compliant Fullstack System - RBAC & Data Isolation

HIPAA · RBAC · Fullstack · REST APIs · Security

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant fullstack system integrating AI with enterprise REST APIs - Python/Flask backend, React.js + TypeScript frontend, Docker containerization; RBAC with authentication and full data isolation ensuring users access only their authorized data; complete audit trail across all data retrieval actions; ~60% reduction in manual retrieval time. Demonstrates the secure fullstack integration pattern Slack's Enterprise Trust team implements for DLP, data retention, and audit log features.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Compliance AI Production Delivery

- Designed and delivered a production compliance AI system (Python, LangGraph, FastAPI) for an external customer - end-to-end ownership: requirements gathering, architecture, secure implementation with audit logging and human-review gates, automated testing, and production deployment in weeks as sole technical owner.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering; translated complex outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Enterprise compliance and security systems** - 100% PHI/HIPAA compliance with Libsodium-encrypted data pipelines; RBAC and access controls across multi-tenant SaaS; full audit logging for every regulated financial and compliance action; zero security incidents across 7yr of production; owned HUD federal compliance pipeline (180,000 domain records, 100K+ housing units) where incorrect data = regulatory violations and incorrect payments.
- **Fullstack production delivery (PHP, Python, React, TypeScript)** - 20+ full-stack modules (PHP backend, Python data pipelines, PostgreSQL, JavaScript/TypeScript, React.js) across 120+ Agile sprints; AWS S3/EC2/RDS/Glue; Docker microservices (50% scalability improvement); RabbitMQ event-driven async with retry logic and idempotency; Redis caching; GitHub Actions CI/CD; 99.5% uptime.
- **Operational excellence and mentoring** - built internal debugging tools reducing triage time 40%; maintained bug count below 10 for 6+ consecutive months; 300+ peer code reviews; mentored 5+ engineers on production practices, security, and code quality; 200+ SQL optimizations (30% performance gain); Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; compliance data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017